

# Generation Interconnection System Impact Study Report

## For Queue Project Q11-0090

Ustick 100 kV  
March 25, 2013

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0090. The Interconnection Customer proposes a 206.8 MW wind facility interconnecting at the Ustick 100 kV substation in the Ohio area. The request was received on December 27, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 78 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$22,634,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

Bowdle Jct - Redfield 100 kV: 156.9% loading (129 MVA) following the outage of the Charles City Rd - Rio Pinar 100 kV; DFAX = 5.3%.

For the loss of Charles City Rd - Rio Pinar 100 kV, the Sour Lake - Bowdle Jct 100 kV loads to 149.2% of its 132 MVA rating. Project DFAX: 5.3%.

Dicks Crk - Orosi 100 kV: 113.4% loading (118 MVA) following the outage of the Orosi - Woodsdale 230 kV L; DFAX = 5.5%.

For the loss of Woodsdale - Orosi 230 kV, the Dicks Crk - Tejas 100 kV L loads to 100.9% of its 133 MVA rating. Project DFAX: 5.5%.

Washington County Cogeneration Facility - Villa Park 230 kV Line: 100.2% loading (589 MVA) following the outage of the Barrett Substation - Wolfe 500 kV; DFAX = 17.1%.

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                          | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------|---------------------------------------------|---------------|-----------------------|
| Redfield - Bowdle Jct 100 kV      | Rebuild line with higher-capacity conductor | \$2,839,000   | 100% (\$2,839,000)    |
| Sour Lake - Bodle Jct 100 kV Line | Rebuild line with higher-capacity conductor | \$7,021,000   | 100% (\$7,021,000)    |
| Dicks Crk - Orosi 100 kV          | Rebuild line with higher-capacity conductor | \$343,000     | 100% (\$343,000)      |

| Facility                                                    | Required Upgrade           | Cost Estimate | Allocation to Project |
|-------------------------------------------------------------|----------------------------|---------------|-----------------------|
| Dicks Creek - Tejas 100 kV                                  | Reconductor line           | \$1,009,000   | 100% (\$1,009,000)    |
| Washington County Cogeneration Facility - Villa Park 230 kV | Reconductor line           | \$13,917,000  | 82% (\$11,422,000)    |
|                                                             | Total allocated to project |               | \$22,634,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.